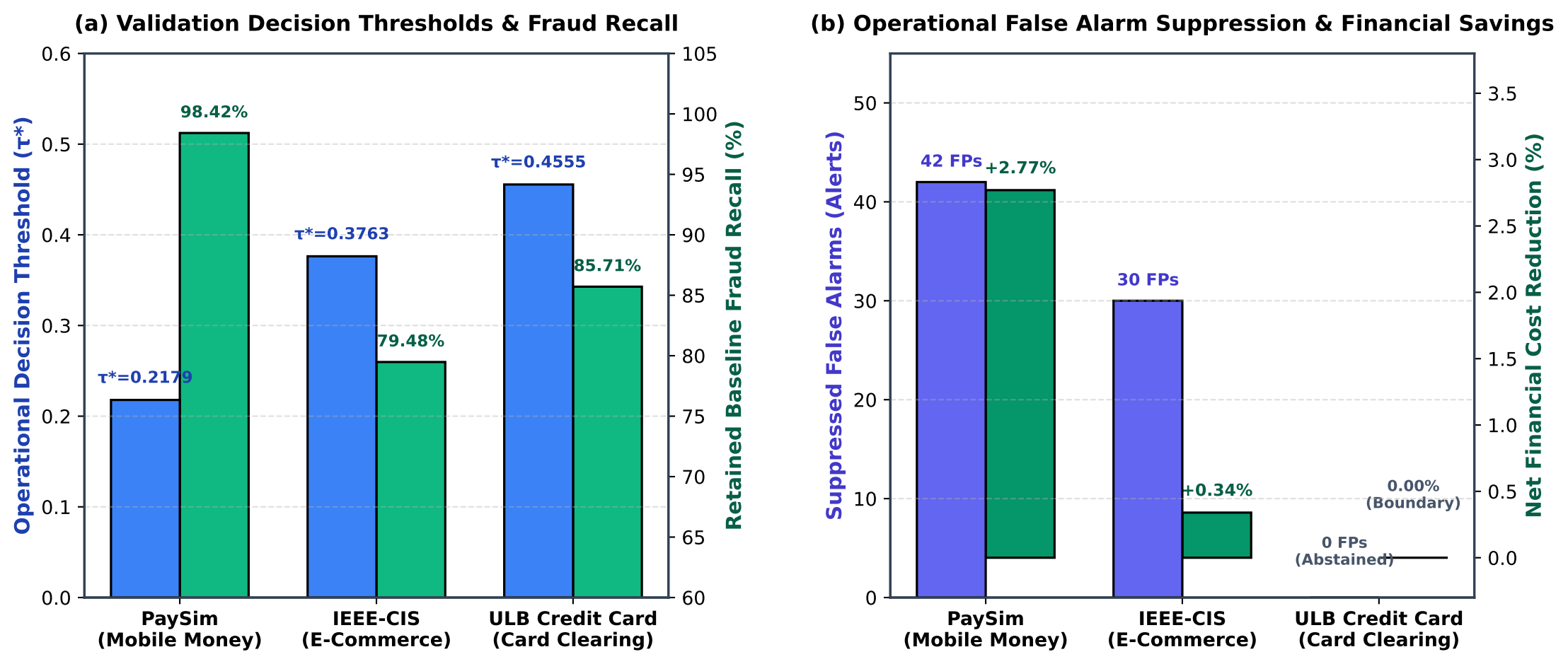
(a) Validation Decision Thresholds & Fraud Recall
Operational Decision Threshold (τ*)
Retained Baseline Fraud Recall (%)
τ*=0.2179
98.42%
τ*=0.3763
79.48%
τ*=0.4555
85.71%
PaySim (Mobile Money)
IEEE-CIS (E-Commerce)
ULB Credit Card (Card Clearing)
(b) Operational False Alarm Suppression & Financial Savings
Suppressed False Alarms (Alerts)
Net Financial Cost Reduction (%)
42 FPs
+2.77%
30 FPs
+0.34%
0 FPs (Abstained)
0.00% (Boundary)
PaySim (Mobile Money)
IEEE-CIS (E-Commerce)
ULB Credit Card (Card Clearing)